

Marblehead → Provincetown — Overnight Race Weather

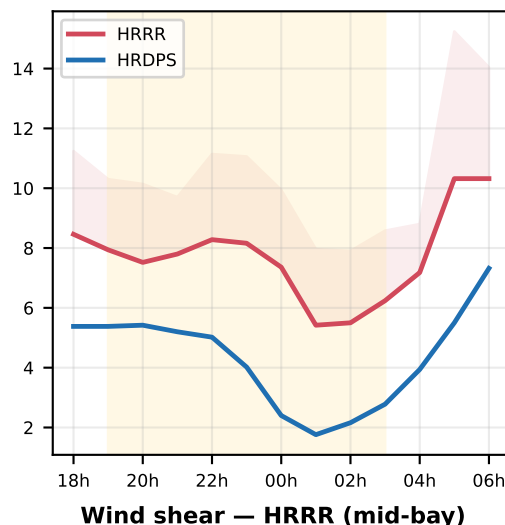
Friday 2026-07-17, 19:00 EDT start · ~03:00 Sat finish · ~40 nm SE across Massachusetts & Cape Cod Bays

Models: NOAA HRRR 3 km + Environment Canada HRDPS 2.5 km (Open-Meteo) · generated 2026-07-16 20:17Z · ~24 h lead — refine race morning

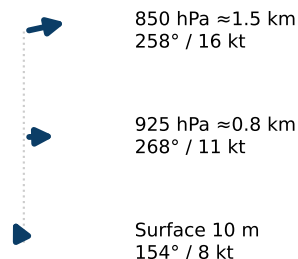
Route & overnight wind (HRRR mean arrows)



Course-mean wind (kt)



Wind shear — HRRR (mid-bay)



~100° veer, wind ~2x in 1 km.
Light SE surface under W-WNW ~16 kt.
925 hPa +1.2°C = inversion (decoupled).

Overnight synopsis

Light, weak-gradient southerly night (flat ~1016 mb, no front). The models AGREE on the overnight: the wind fills S/SW 8-12 kt offshore, a starboard-tack reach that gradually frees toward Race Point, seas ≤2 ft. They SHARPLY DISAGREE on the start: NOAA HRRR has SE ~10 kt (a reach/beat down the rhumb) while Canada HRDPS shows a NW land-breeze ~7 kt (offshore), dying to near-calm by ~01:00 before the southerly fills — opposite first-leg angles. HRDPS has verified better locally this season, but at a post-sunset transition neither is reliable: sail the breeze that's actually on the water, then get offshore into the steadier southerly. Hour-by-hour, tactics, tide and hazards overleaf.

Conditions at a glance

Gusts — steady, modest

HRRR gust factor ~1.4-1.5, peaks ~15-16 kt over ~9 kt mean; gustiest mid-bay/Race Pt pre-dawn. HRDPS gives no gust field. CAPE 0 + only ~11-15 kt at 925/850 hPa → no convective or momentum-driven gusts (unlike a daytime sea-breeze). Plan smooth, not puffy.

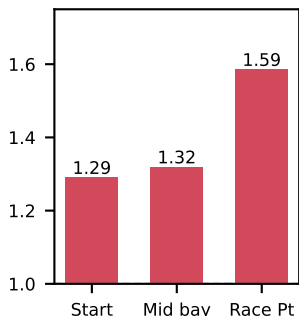
Sea state — benign

~0.2 m / 0.7 ft short-period wind chop (NWS ANZ231/233 seas ≤1-2 ft). SST ~21°C / 69°F. Flat water — the limiter is wind, not waves.

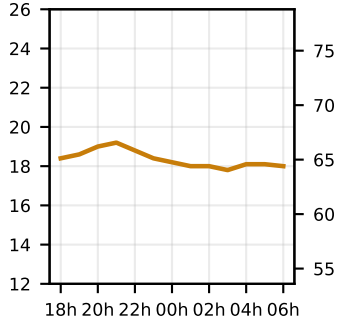
Temperature — mild & near-steady on the water

~18°C / 64-65°F all night on the course; the sea moderates it (drops only ~1°F offshore; up to ~10°F near the P-town shore). Warm water under cooler air keeps it damp → reinforces the fog watch. Layers for damp, not cold.

Gust factor (HRRR)



Air temp on the water (°C)

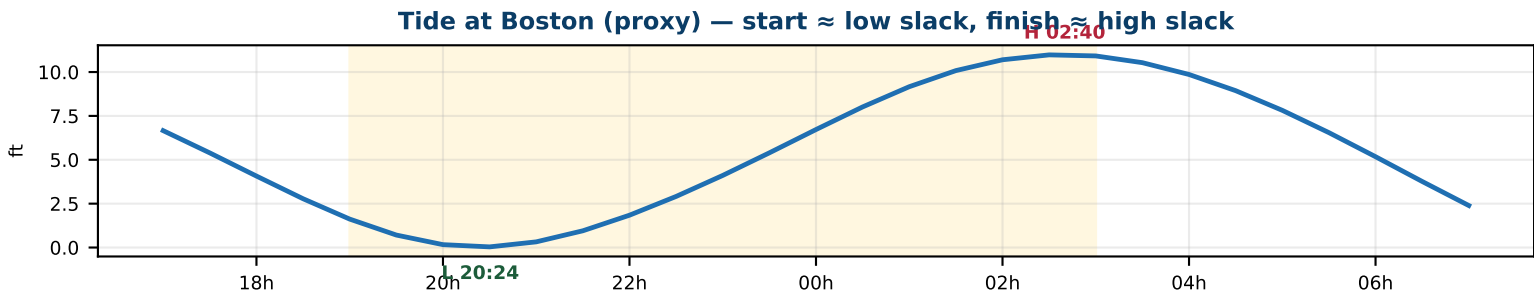


Tactics, Strategy & Timing

Marblehead → Provincetown · Fri 2026-07-17 overnight

Corridor wind — dir°/kt, H=HRRR (top) / C=HRDPS (bottom)

Time ET	Start (Tinkers Rk)	Off Nahant	Mid Mass Bay	Stellwagen edge	Race Pt / P-town
19:00 H C	136/10 285/8	139/10 268/7	137/10 183/4	143/7 164/5	123/3 161/3
21:00 H C	139/8 309/6	148/9 311/6	146/9 230/2	146/7 180/5	156/6 192/6
23:00 H C	153/8 321/4	156/9 333/4	155/7 319/1	162/9 212/4	158/7 213/8
01:00 H C	233/2 268/1	169/5 127/1	150/6 163/1	143/8 204/2	147/6 221/4
03:00 H C	081/4 155/3	142/8 160/4	153/8 169/3	166/6 175/2	165/5 203/2
05:00 H C	180/9 190/5	180/9 186/6	190/11 176/5	192/12 166/5	194/10 180/6



Tactics & strategy

① Start (19:00-21:00) — models disagree, so read the water

HRRR: SE ~10 kt (reach/close-hauled). HRDPS: NW land-breeze ~7 kt (offshore), dying to calm ~01:00. Opposite angles — don't pre-commit. Sail the actual dock breeze; whatever it is it's LIGHT (≤8 kt). Get clear air off the line, then work OFFSHORE — the Marblehead shore is the light trap.

② The crossing (21:00-01:00) — reach in the filling southerly

Both models fill S/SW 8-12 kt offshore — a starboard reach that frees as the wind veers right. Pressure lives offshore/south (mid-bay ~10 kt vs shore 2-6). Commit south; stay on the pressure side, don't over-stand north toward Cape Ann.

③ ⚠️ LNG Northeast Gateway — hard avoid

Two STL buoys at ~42.40 N (−70.60/−70.59) sit just N of the direct line. NO ENTRY within 500 m, no anchoring within ~1 nm. The rhumb passes ~5 nm south — staying on/south of the line is both faster (more breeze) and legal.

④ Race Point & finish (01:00-03:00) — the tide gate

S/SSW 8-11 kt, gusts to 15 into Cape Cod Bay. Flood builds to Boston high 02:45 ≈ finish. Round Race Point BEFORE ~02:45 to carry the flood in; later boats meet the building ebb. Finish current itself is minor (near high slack).

⑤ Wind shear — expect shifty, veered puffs

Light SE/S surface sits under a W-WNW 13-16 kt flow at 925/850 hPa with a low-level inversion (decoupled night). Puffs mix a little of that down — shifter and veered toward W/NW vs the mean, but capped ~16 kt. Toward dawn the layer mixes and the surface veers/builds to the gradient. Don't bank on one wind angle; sail conservatively through puffs, keep the boat in pressure.

⑥ Night & hazards

Dark — thin crescent sets early, so nav by instruments, watches + lights set. Fog watch: July S-flow over ~20°C water; NWS not flagging Fri night but recheck AM. Seas ≤2 ft, no SCA expected.

Confidence: moderate on the overnight reach + tide timing (models agree); low on the exact start (transition + lead). Re-run race morning + pre-start.